

Exact weighted monochromatic-K4 certificates with a pending prior-bound comparison

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Judging and evaluation record: the submissions discussed in this document have been judged and evaluated by Alejandro Zarzuelo Urdiales for OpenMath 2026. This dated review records the stated verification, classification and unresolved holds; it is a preliminary evaluation rather than a signed award decision.

Event provenance: OpenMath 2026 ran 27 September-2 October 2026 with worldwide online participation. Detailed organizer listings specify Harvard Innovation Labs for the 27 September in-person event; the OpenMath programme advertises a hybrid kickoff at MIT. Sundai identifies its Harvard/MIT community. Organizer sources:

<https://rsihouse.ai/openmath> ; <https://luma.com/yzp9abvr> ;

<https://www.sundai.club/events/boston/recursive-learning-hack-with-harvard-innovation-labs>

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Central source and adjudication archive: <https://github.com/alejandroz/alejandrozu/openmath-2026-judging>

Abstract

A1024-block binary template and positive integer weights give exact weighted monochromatic-K4 density $200137750014677779172692777477/6640289795261907019148161833841 \approx 0.030139911990$. Independent exact arithmetic confirms the number. A stronger pre-event Technion announcement blocks an unqualified new-record claim; method and formalization novelty are to be assessed separately.

$c_4 \leq 200137750014677779172692777477/6640289795261907019148161833841$ under the specified weighted-template lifting convention; numerical record status held.

Finite weighted construction and lifting

A binary symmetric template on 1024 blocks is blown up using positive integer weights. The weighted ordered four-vertex probability includes repeated block choices under the specified loop convention. The exact numerator and denominator are $200137750014677779172692777477$ and $6640289795261907019148161833841$, giving density approximately 0.030139911990 . A finite count alone does not establish an asymptotic extremal bound; the generic lifting theorem supplies graph families whose densities approach the weighted objective.

Independent exact arithmetic reproduced the selected weighted number using modular counts and a Chinese-remainder reconstruction. The JSON data were compared with the Lean literals. Those numerical checks and the formal finite-to-limit theorem are different parts of the evidence. The Sakana packet evaluates the concrete count outside Lean under its submitted exact-checker protocol; the HTPeo packet also supplies a kernel count route. The common verification register must report exactly which closure and checker were freshly executed.

The pre-event comparison

The result is below the frozen McKay benchmark approximately 0.030142273432. However, Technion's official January 2026 announcement states $c_4 < 0.030139$, which is stronger than 0.030139911990. The announcement establishes a specific priority lead, although it does not expose the complete prior certificate in the inspected page.

Until the earlier construction and conventions are compared, this paper must not claim a new numerical record. Correct exact construction and novelty are independent questions. A genuinely different construction method, reusable certificate or first faithful formalization may still be publishable, but that delta needs its own comparison and cannot be inferred from beating an obsolete benchmark.

Concrete release gate

Request the earlier dated theorem and certificate from the Technion authors, compare the density normalization and target class, and obtain an independent specialist opinion. Then decide whether this is original construction mathematics, useful formalization of a known technique, or a reproduction below the frontier. Independent competition development can matter to contribution attribution without changing the chronological best-known bound.

This held manuscript remains separate from the original-results anthology. Its final version should include the template/weight data, exact counting specification and lifting theorem, credit the prior bound and avoid transforming a small decimal gain into a broad Ramsey-theory claim.

Verification and reproducibility

Dated partial fresh replay: the qualification on 5 October 2026 at 19:50:13 UTC verifies 2083 of 2103 original source modules. All 2048 independent certificate kernels and 35 common providers passed with unchanged scientific bodies; imported dependencies were reused through a separately qualified official-header route, while the new kernel bodies were checked again. Nine symmetry-support sources and eleven whole-import aggregate sources remain unrun. Both the original 24-request project audit and the separate 24-request supplement remain unrun, with zero project selected prints. Two earlier commit-reserve interruptions are preserved as resource attempts, not mathematical failures; later solo source replay completed the remaining kernel scope. Six isolated cache-route axiom prints remain separate evidence and do not qualify the main project endpoints. Full-project and selected-endpoint verification, numerical priority, score, award and release remain unqualified. Scope record: Verification/root-HT-actual2083-two-leases-all-kernel-sources-partial-readonly-qualification-20261005T195013776591Z.json. Immutable checkpoint: Verification/ht-ramsey-actual2083-all-kernel-sources-immutable-20261005T194825772916Z-0f76ae05.json.

The preserved contribution record and source locator below identify the selected certificate. No new endpoint statement is invented for this editorial manuscript.

Sources and attribution: 1. [math.technion.ac.il / noam-feinstein](http://math.technion.ac.il/~noam-feinstein/); 2. arXiv source: 2206.04036; 3. [app.autolab.ai / clique-cluster-ramsey-multiplicity](http://app.autolab.ai/); 4. [alejandrozou/openmath-2026-judging / E08-RAMSEY.txt](https://alejandrozou/openmath-2026-judging/E08-RAMSEY.txt); Central source and adjudication archive.